

SECTION B

Answer all questions

Read through the following article carefully.

Freely adapted from:

The Physics of Optical Fibres
By Justino Luis Moreno

Paragraph

Basically, an optical fibre is just a piece of glass along which you send light. A similar procedure was first demonstrated not with glass but with water flowing from a spout in 1840 by scientists Daniel Colladon and Jacques Babinet. Babinet later published his work in an article entitled “On the reflections of a ray of light inside a parabolic liquid stream”. This effect can be reproduced quite easily in a school lab using the apparatus shown in Figure 1.

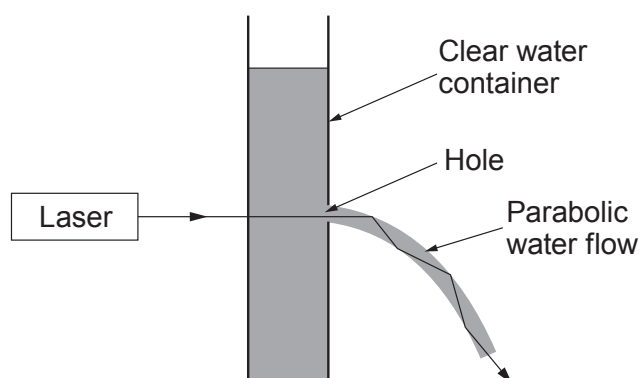


Figure 1

Although this set up is pleasing to the eye and is the basis of some water features, it isn't much use for international telecommunication! A standard optical fibre is shown in Figure 2 along with a ray of light entering it.

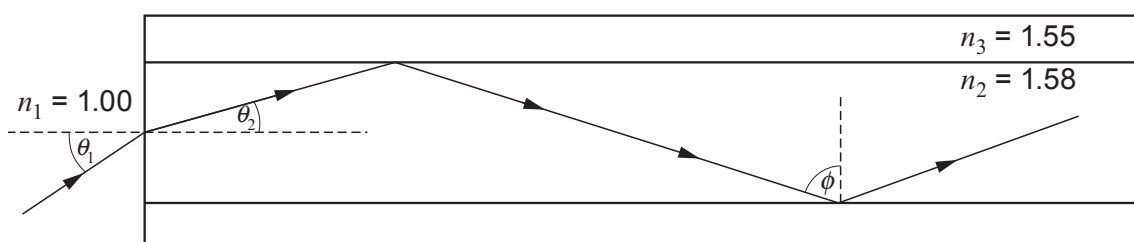


Figure 2

This ray of light is repeatedly reflected along the length of the optical fibre. If the entrance angle (θ_1) is small enough then an effect called total internal reflection (TIR) means that the light is completely reflected each time resulting in no light escaping as it travels along the optical fibre. The physicists in charge of designing these things can prove that the minimum value angle ϕ can have for TIR to take place is given by the equation:

$$\phi = \sin^{-1}\left(\frac{n_3}{n_2}\right)$$

Paragraph

This angle is around 80° for the optical fibre shown. The minimum angle for ϕ means that there is a maximum angle for θ_1 . This gives an acceptance cone where the input light gets propagated without loss. A quick bit of geometry shows you that the exit angle is the same as the entry angle so that when light exits it produces an almost perfect cone shaped beam with a circular cross-section as can be seen in Figure 3.

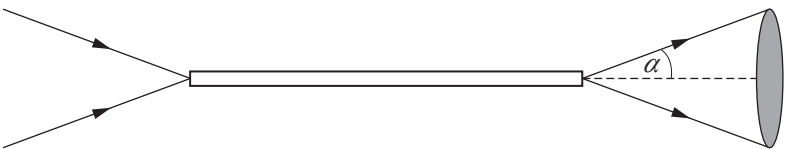


Figure 3

An important term in optical fibre technology is maximum bit rate. Since each pulse represents a bit of data, this is the highest frequency of pulses that can be sent down an optical fibre before the pulses start to overlap and become indistinguishable from each other. A monomode optical fibre cable of length 80km can comfortably have a maximum bit rate of 10 Gbs^{-1} . This means that 10^{10} pulses can pass along its length every second without overlapping. A typical telephone conversation requires around 10 kbs^{-1} meaning that one monomode optical fibre cable can carry a million telephone conversations simultaneously. High definition TV requires a much higher bit rate and a 10 Gbs^{-1} fibre will only handle around 2 000 high definition TV signals.

One of the most important factors that limit data transfer in optical fibres is multimode dispersion. This, put simply, is to do with the entrance cone of light in Figure 3. There is a range of distances that the pulses have to travel because there are a variety of angles at which they can travel along the fibre. The pulses then become spread out and indistinct, ruining the digital signal. Multimode dispersion is eliminated by using monomode optical fibre cables which have very thin cores. As a rule, monomode fibres have a core diameter of around $8\mu\text{m}$. This means that the core is less than 10 wavelengths thick so that the light stops behaving like rays. For monomode fibres there is only one propagation direction – along the axis.

Another drawback of sending signals down long lengths of optical fibres is that some of the light is either scattered or absorbed by the glass molecules themselves (an effect known as attenuation). Although no light escapes the fibre due to TIR there are other losses involved and these are usually summarised by using a decibel (dB) scale. This scale is defined by saying that a 10 dB decrease (-10 dB) in power is when the power has dropped to 10% of its input value. A loss of -20 dB then corresponds to a drop to 1% power and -30 dB is a drop to 0.1% power. The following table shows the relationship between dB values and power ratio:

dB	Power ratio $\left(\frac{P}{P_0}\right)$
-5	0.316
-10	0.100
-15	0.032
-20	0.010

Table 1

The losses of optical fibres are usually quoted in the unit dB/km and some modern optical fibres can have values as low as 0.01 dB/km. Hence, each km of cable loses 0.01 dB meaning that you can use 1 000km before your signal is down to 10% strength. Optical fibres have come a long way since they were born in a fountain of light nearly 200 years ago. They stretch out (literally) to all areas of the world bringing light, sound and broadband wherever they go. Technical advances mean that data can be sent at a rate of 1.05×10^{15} pulses per second over a distance of 50 km with only one monomode optical fibre. Nonetheless, the technology has its limitations of which attenuation and multimode dispersion are but two.

Examiner
only

8. Answer the following questions in your own words. Direct quotes from the original article will not be awarded marks.

(a) Explain why the water in Figure 1 flows in a downward curve (paragraph 1). [2]

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(b) Show that the minimum angle (ϕ) for total internal reflection in the optical fibre of Figure 2 is less than 80° (paragraph 3). [2]

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(c) Explain why the light emerging from the optical fibre is in the shape of a cone of light (paragraphs 3 & 4 and Figures 2 & 3). [2]

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(d) Calculate the angle (α) of the cone of light emerging from the optical fibre in Figure 3 (see paragraphs 3 & 4 and Figures 2 & 3). [4]

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- (e) Rhys makes the following claim:

“A multimode optical fibre that can transfer data at a maximum bit rate of 500 Mb s^{-1} over a distance of 1 km would be able to transfer data at a maximum rate of 50 Mb s^{-1} over a distance of 10 km.”

Discuss whether or not Rhys's claim is valid (paragraph 5 & 6).

[3]

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- (f) A certain type of optical fibre has an attenuation of 0.8 dB/km . When the signal decreases to 6% of its original intensity it must be amplified or the signal will be lost. An engineer intends to install these optical fibres in lengths of 20 km before each amplifier. Determine whether or not this is an appropriate length of optical fibre to use (see paragraph 7 and Table 1). [3]

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- (g) Aled claims that a core diameter of around $13 \mu\text{m}$ is thin enough for a monomode optical fibre because a wavelength in air of $1.3 \mu\text{m}$ is usually used in communications. Rhian claims that this is nonsense because the wavelength in the optical fibre is changed due to its refractive index. Justify by including a calculation who is correct (paragraph 6). [4]

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END OF PAPER